

The role of tongue root anteriority in the Dolgan vowel system

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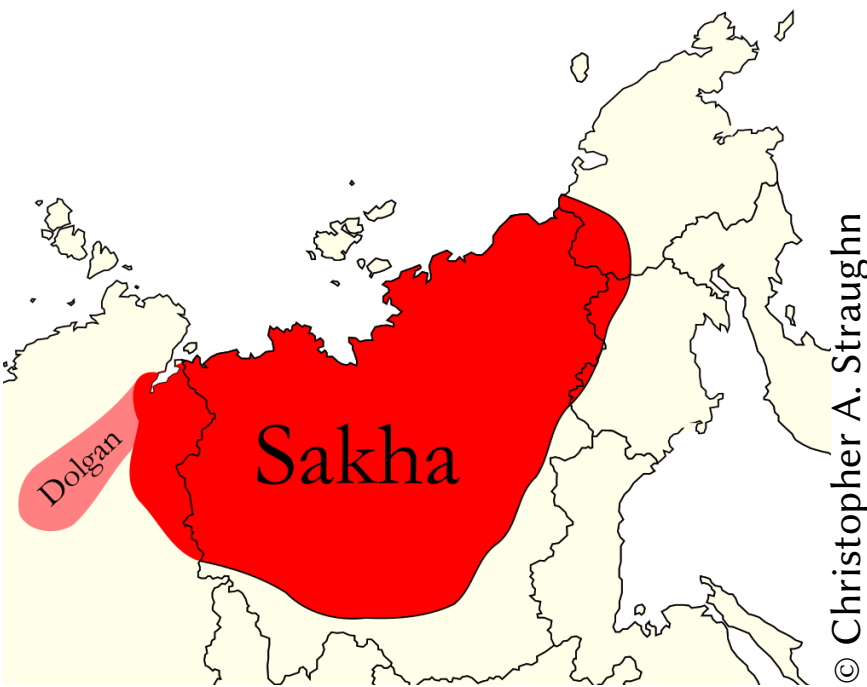
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Dolgan

- Lena Turkic language, spoken on Taymyr peninsula (Arctic Siberia)
- Endangered, ~5k speakers, loss of intergenerational transmission

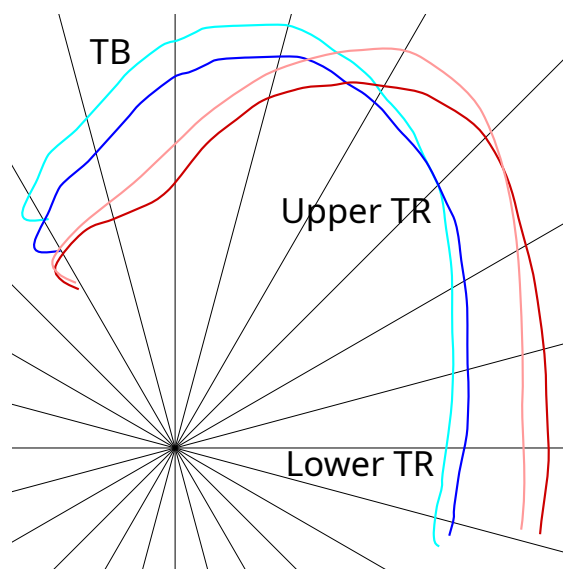


Phonological evidence: alternation of /G/

/G/: surfaces as **velar** or **uvular** depending on preceding vowel

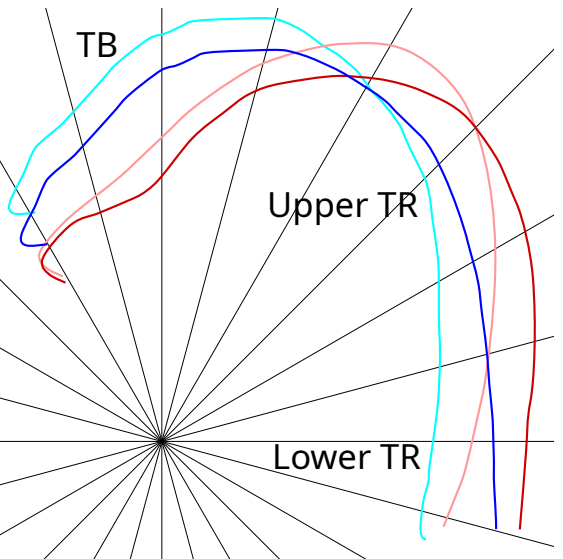
Kyrgyz (&c.)	front		back	
	high	low	high	low
high	[ini-ye]	[a:ɪw-ɪa]	‘brother-DAT’	‘bee-DAT’
low	[ene-ye]	[ɑɑ-ɪa]	‘mother-DAT’	‘saw-DAT’

- **uvular G** / V_[+back] → **velar G** / V_[+back] →
- ↳ V backness associated with TR backness
- coproduced: TB backness & TR backness
- (Washington, 2016, 2019)



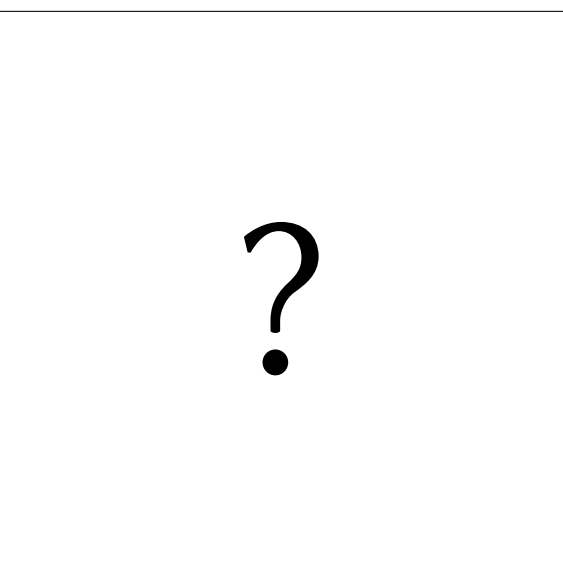
Sakha	front		back	
	high	low	high	low
high	[iti:-gɛ]	[ɑɪw:-gɑ]	‘heat-DAT’	‘century-DAT’
low	[yjɛ-ɪɛ]	[tabɑ-ɪa]	‘century-DAT’	‘reindeer-DAT’

- **uvular G** / V_[+high] → **velar G** / V_[+high] →
- ↳ V height associated with TR backness
- coproduced: TB height & TR backness
- (Vaux, 2009)

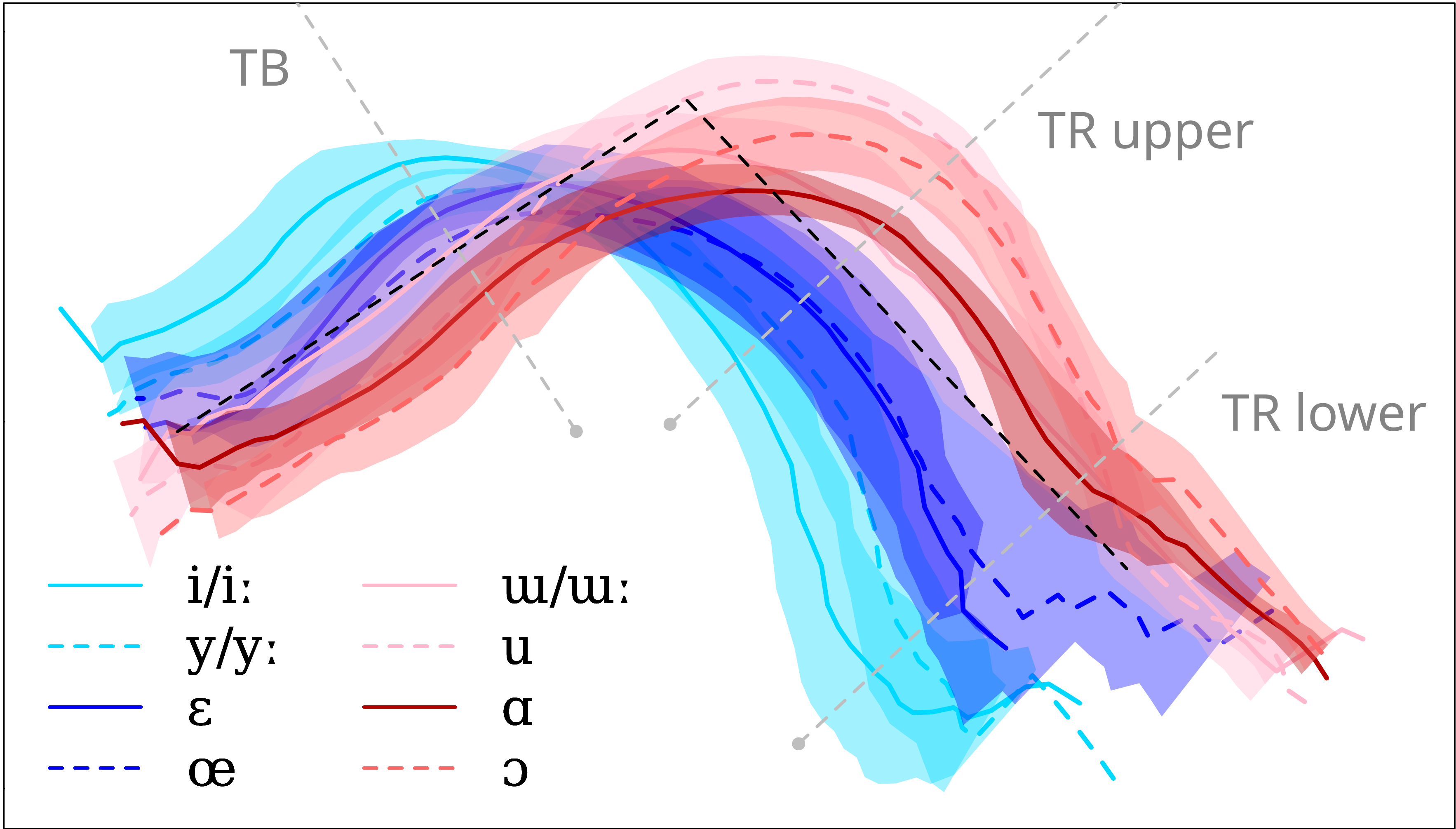


Dolgan	front		back	
	high	low	high	low
high	[iti:-gɛ]	[ɑɪw:-gɑ]	‘heat-DAT’	‘century-DAT’
low	[yjɛ-gɛ]	[tabɑ-ɪa]	‘time-DAT’	‘reindeer-DAT’

- **uvular G** / V_[+hi,+bk] → **velar G** / elsewhere
- (Убратова, 1985; Däbritz, 2022)
- ↳ V height & backness associated w. TR backness
- coproduced: TB height & TB & TR backness!



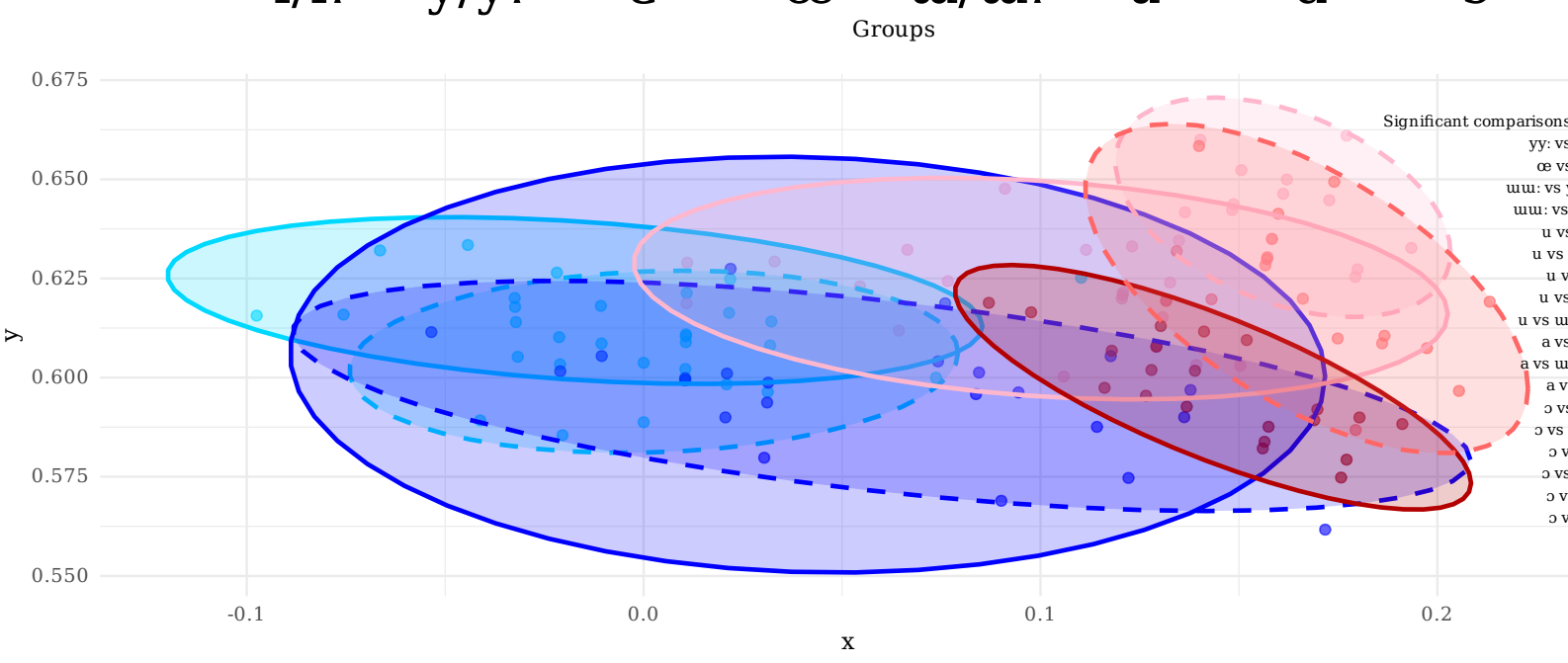
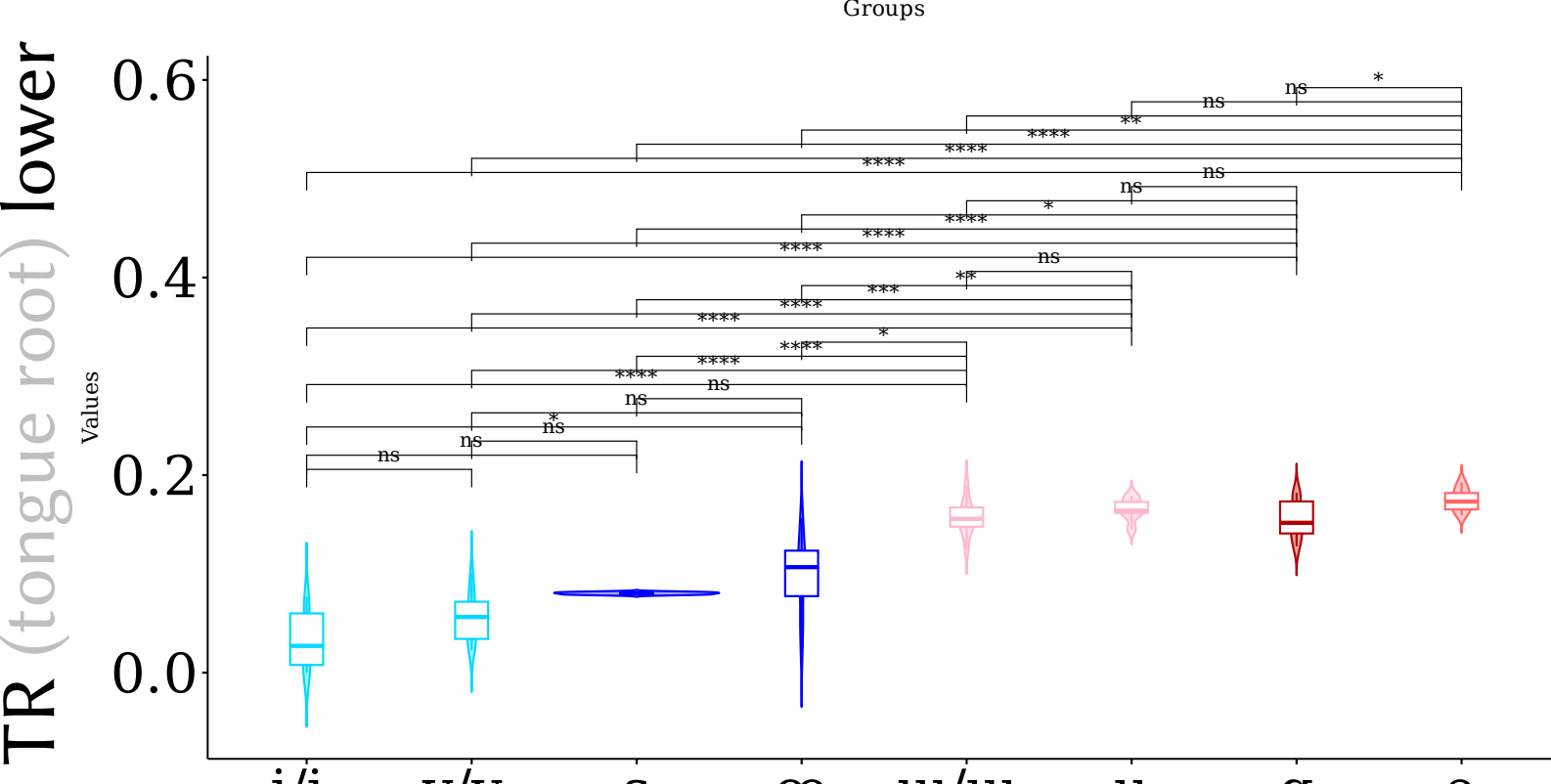
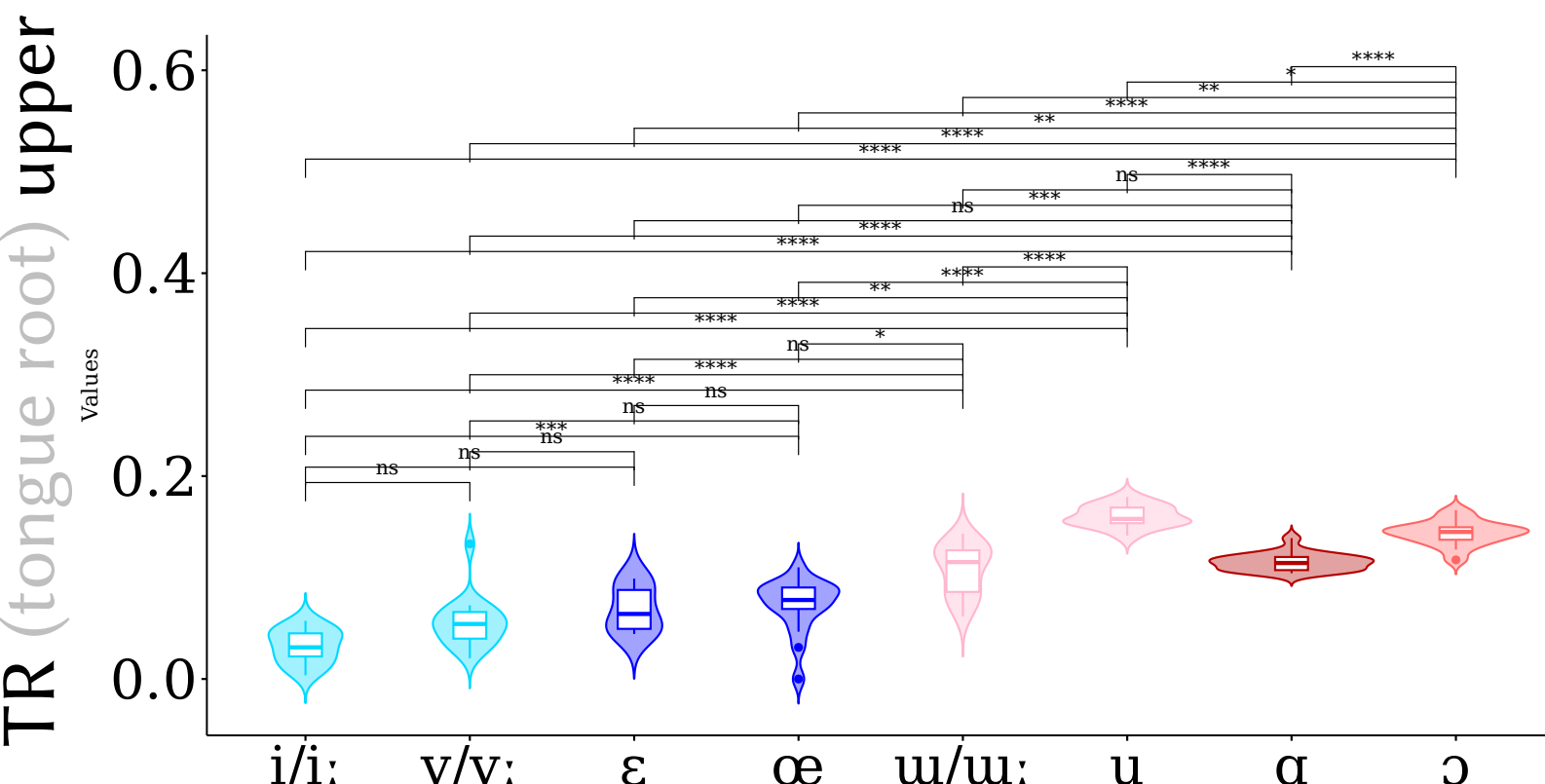
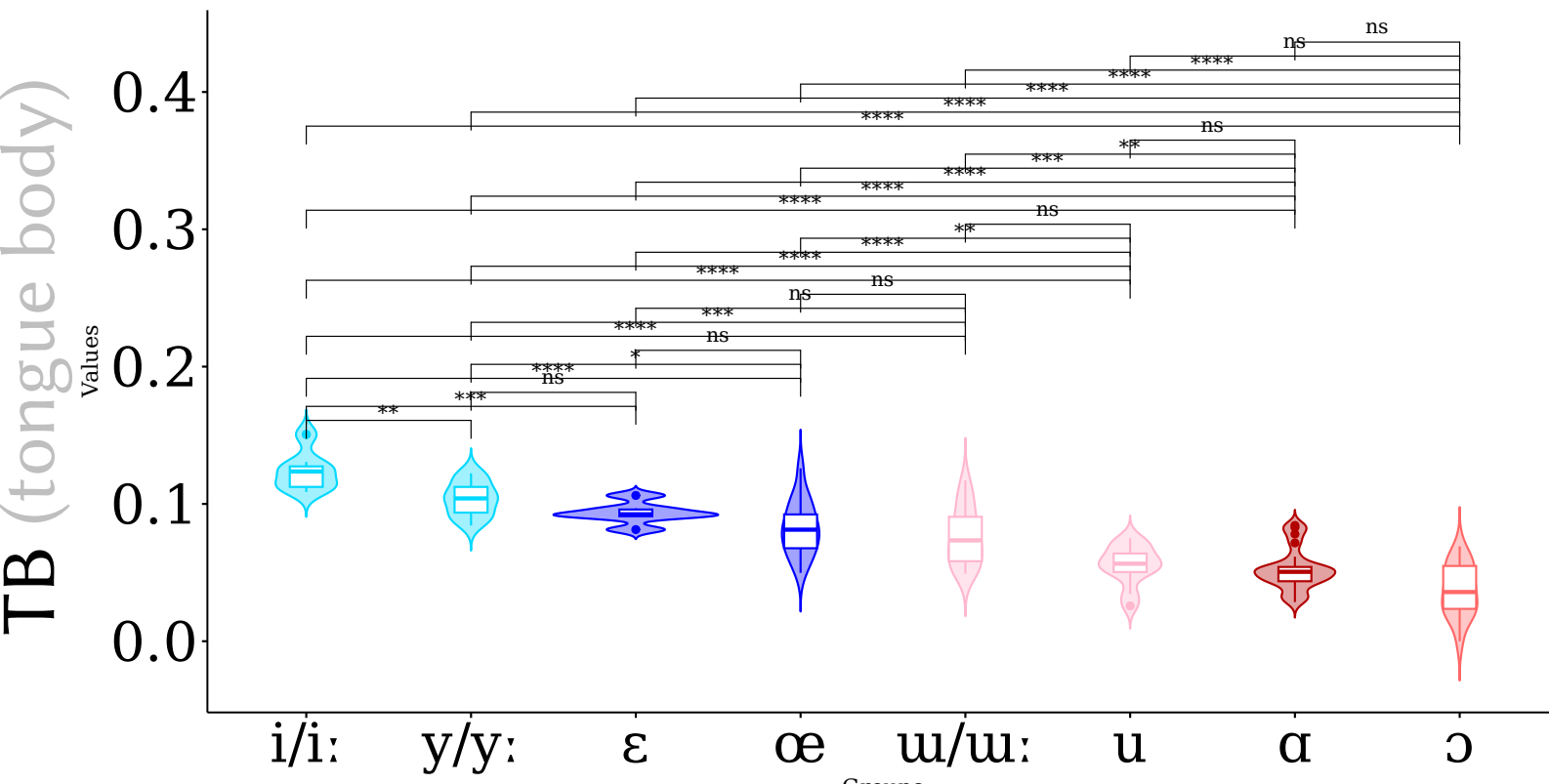
Results



← tongue tip left

The **tongue root** plays unexpected roles in the **Dolgan** vowel system

Significance (paired t-tests)



	pair	TB		TR	
		middle	PMC	upper	lower
height	i-ɛ	0.00052	1.00000	0.28013	0.09608
	y-œ	0.02035	1.00000	1.00000	0.19771
	u-ɑ	0.00555	0.00049	1.00000	1.00000
	u-ɔ	0.16815	0.01150	0.02140	1.00000
rounding	i-u	1.6e-07	0.20473	3.5e-08	2.4e-05
	y-u	1.3e-07	4.3e-13	3.8e-13	1.2e-07
	ɛ-ɑ	5.0e-06	1.00000	0.11347	1.8e-09
	œ-ɔ	2.5e-06	4.2e-10	2.7e-08	0.00252
backness	i-y	0.00732	0.00584	0.12860	1.00000
	ɛ-œ	1.00000	1.00000	1.00000	1.00000
	ɔ-ɑ	0.27664	1.3e-06	4.3e-07	0.03711
	u-u	0.07095	1.4e-06	1.4e-05	1.00000

p values: ns ≥ 0.05 > * ≥ 0.01 > ** ≥ 0.001 > *** ≥ 0.0001 > ****

Methodology

Recorded L1 speaker of Lower Xatanga Dolgan (author 4)

- Wordlist task, carrier phrase *Зоя __ диир* ‘Zoya says __’ (in custom orthography)
- Articulate Instruments’ Micro ultrasound system, AAA software (Wrench, 2017)

Annotated midsagittal tongue surface using UltraTrace (Murphy et al., 2020)

- frame closest to midpoint of V in 1st syllable of each word

Plotted and **analysed** using ultrapolaRplot (Outkin & Washington, 2024)

- Perpendicular rays intersect various areas of the tongue (mid TB, upper TR, etc.)
- Pairwise t-test at intersections for significance (adjusted for Type I error)
- Pairwise t-test compares maximum curvature (PMC) groups by calculating differences in distance to origin [future work: point on best fit line of curvatures]

Main findings

- V height implemented via TB height (**expected**)
- V height not implemented via TR backness (**unexpected**)
- V backness implemented via TB height (**unexpected?**)
- V backness implemented via TR backness (**expected**)
- V roundedness implemented via TB and TR (**unexpected**)

Impressionistically:

- 5 levels of TR backness: i > y > ɛ/œ > ɑ/u > ɔ/u

Quantitative methodology and impressionistic methodology not perfect match

References

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poster



<https://github.com/SwatPhonLab/ultrapolaRplot>

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